

Computer Architecture and Organization Lab

Mid Exam T1.2024-2025

Solved By : Laith Q. Amro

هذا الملف تم حلّه من قبل الطالب : ليث عمرو

هذه الحلول هي جهد بشري قابل للخطأ،
إن أصبت فمن الله، وإن أخطأت فمَنّي ومن الشيطان.

هذا العمل مقدّم ثوابه لروح والدي (قاسم مفيد عمرو)
فمن وجدَ فيه فائدة فليدعُ له بالرحمة والمغفرة.

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Problem #1 : Choose the correct answer [5/6pts]

1. Component instantiation is the part of _____ modeling.
A) Behavior B) Component C) Dataflow **D) Structural**
2. Which logic function is described in the VHDL following code?

```
ARCHITECTURE my_func OF my_logic IS
begin
  process(a, b, y)
  begin
    IF(a = '0' and b = '0') THEN
      y <= '0';
    ELSIF (a = '1' and b = '1') THEN
      y <= '0';
    ELSE
      y <= '1';
    END IF;
  END process;
END my_func;
```

- A) AND **B) XOR** C) OR D) XNOR
3. Using VHDL _____, one can transfer data between components or inside them.
A) Port B) Package C) Component **D) Signal**
 4. Generics are declared in the _____ declaration part of a VHDL design.
A) Port declaration B) Component
C) Entity D) Process
 5. Multi-bit digital values are represented using _____.
A) Standard Logic Signals (std_logic)
B) Standard Logic Vectors (std_logic_vector)
C) Integer Signals
D) Boolean Signals

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6. Which logic function is described in the VHDL following code?

```
ARCHITECTURE my_func OF my_logic IS
begin
  process(a, b, y)
  begin
    IF(a = '0' and b = '1') THEN
      y <= '0';
    ELSIF (a = '1' and b = '0') THEN
      y <= '0';
    ELSE
      y <= '1';
    END IF;
  END process;
END my_func;
```

A) AND

B) XOR

C) OR

D) XNOR

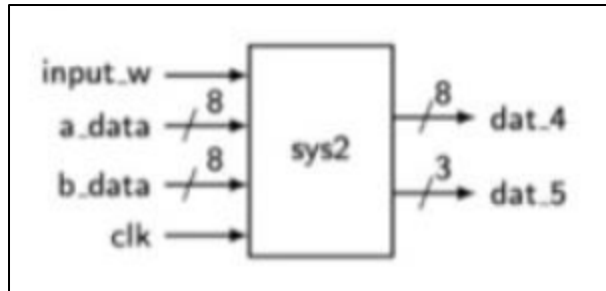
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Problem # 2: Write the VHDL entity declaration that describes the following black-box diagram in the below table. [5pts]



entity sys2 is

port (

input_w, clk : in std_logic;

a_data, b_data : in std_logic_vector(7 downto 0);

dat_4 : out std_logic_vector(7 downto 0);

dat_5 : out std_logic_vector(2 downto 0));

end sys2;

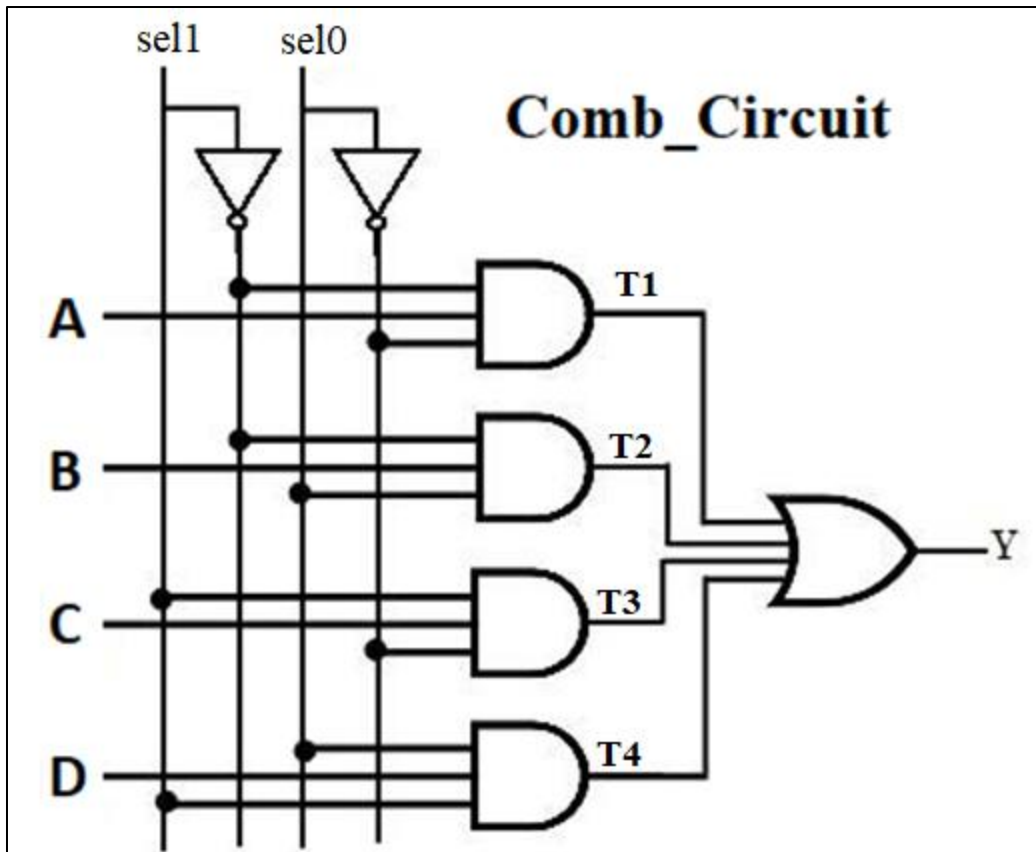
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Problem # 3: Based on the combinational circuit called **Comb_Circuit**, answer the following two questions. [10pts]



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A) Complete the below **Dataflow** model VHDL code for **Comb_Circuit**, Use the signals T1, T2, T3, and T4. [5pts]

```
library IEEE;
use IEEE.std_logic_1164.all;

entity Comb_Circuit is
port (
    A, B, C, D : in std_logic;
    sel : in std_logic_vector(1 downto 0);
    Y : out std_logic
);
end Comb_Circuit;

architecture Comb_Circuit_DF of Comb_Circuit is
    signal T1, T2, T3, T4 : std_logic;
begin
    T1 <= A and not sel(1) and not sel(0);
    T2 <= B and not sel(1) and sel(0);
    T3 <= C and sel(1) and not sel(0);
    T4 <= D and sel(1) and sel(0);

    Y <= T1 or T2 or T3 or T4;
end Comb_Circuit_DF;
```

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B) Complete the **Behavioral** model VHDL code for **Comb_Circuit** using conditional signal assignment statement. [5pts]

```
Library IEEE;
use IEEE.std_logic_1164.all;

entity Comb_Circuit is
port ( A, B, C, D : in std_logic;
      sel : in std_logic_vector(1 downto 0);
      Y : out std_logic
);
end Comb_Circuit

Architecture Comb_Circuit_behave for Comb_Circuit is
begin

process(A,B,C,D,sel)
begin

Y <= A when sel="00" else
    B when sel="01" else
    C when sel="10" else
    D when sel="11" else
    'Z' when others;

end process;
end Comb_Circuit_behave;
```

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Problem # 4: Assuming the VHDL files shown in the given appendix, answer the following two questions. [10pts]

- Appendix

NAND_3_input_gate.vhd

```
library IEEE;
use IEEE.std_logic_1164.all;

entity NAND_3_input_gate is
port(
    inp1: in std_logic;
    inp2: in std_logic;
    inp3: in std_logic;
    out1: out std_logic
);
end NAND_3_input_gate;

architecture behav_NAND_3_input_gate of NAND_3_input_gate is
begin
    out1 <= not (inp1 and inp2 and inp3);
end behav_NAND_3_input_gate;
```

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OR_2_input_gate.vhd

```
library IEEE;
use IEEE.std_logic_1164.all;

entity OR_2_input_gate is
port(
    inp1: in std_logic;
    inp2: in std_logic;
    out1: out std_logic
);
end OR_2_input_gate;

architecture behav_OR_2_input_gate of OR_2_input_gate is
begin
    out1 <= inp1 or inp2;
end behav_OR_2_input_gate;
```

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XOR_2_input_gate.vhd

```
library IEEE;
use IEEE.std_logic_1164.all;

entity XOR_2_input_gate is
port(
    inp1: in std_logic;
    inp2: in std_logic;
    out1: out std_logic
);
end XOR_2_input_gate;

architecture behav_XOR_2_input_gate of XOR_2_input_gate is
begin
    out1 <= inp1 xor inp2;
end behav_XOR_2_input_gate;
```

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MUX_2x1.vhd

```
library IEEE;
use IEEE.std_logic_1164.all;

entity MUX_2x1 is
port(
    inp1: in std_logic;
    inp2: in std_logic;
    sel: in std_logic;
    out1: out std_logic
);
end MUX_2x1;

architecture behav_MUX_2x1 of MUX_2x1 is
begin
    out1 <= (not sel and inp1) or (sel and inp2);
end behav_MUX_2x1;
```

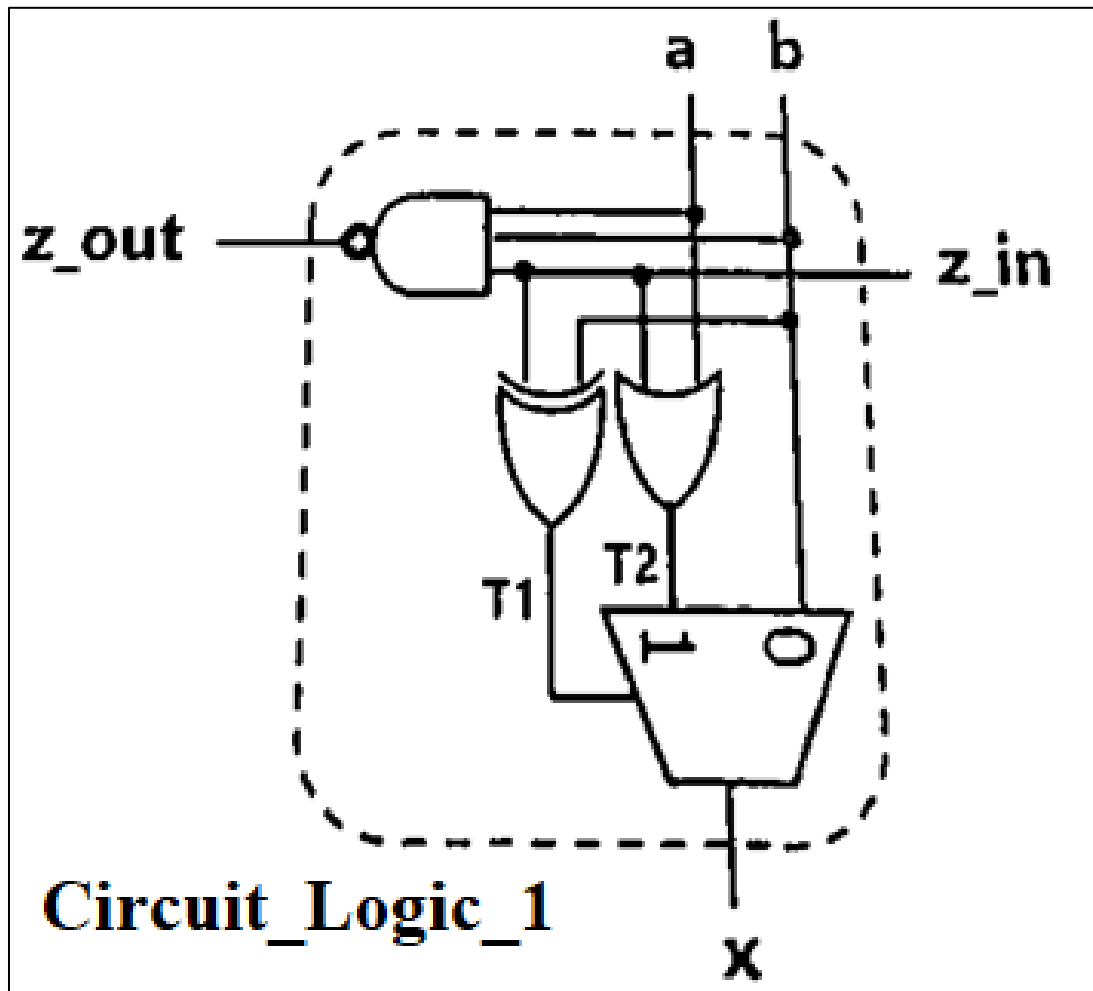
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A) Write the **Structural** model VHDL code for the **Circuit_Logig_1** in the table below using Positional association. [5pts]



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Library IEEE;

use IEEE.std_logic_1164.all;

entity Circuit_Logic_1 is

**port (a, b, z_in : in std_logic;
 z_out, x : out std_logic);**

end Circuit_Logic_1;

Architecture struct_Circuit_logic_1 of Circuit_logic_1 is

**component NAND_3_input_gate port (inp1, inp2, inp3 : in std_logic;
 out1 : out std_logic); end component;**

**component OR_2_input_gate port (inp1, inp2 : in std_logic;
 out1 : out std_logic); end component;**

**component XOR_2_input_gate port (inp1, inp2 : in std_logic;
 out1 : out std_logic); end component;**

**component MUX_2x1 port (inp1, inp2, sel : in std_logic;
 out1 : out std_logic); end component;**

signal T1,T2 : std_logic;

begin

NAND1: NAND_3_input_gate port map (a, b, z_in, z_out);

OR1: OR_2_input_gate port map (z_in, a, T2);

XOR1: XOR_2_input_gate port map (z_in, b, T1);

MUX1: MUX_2x1 port map (b, T2, T1, X);

end struct_Circuit_logic_1;

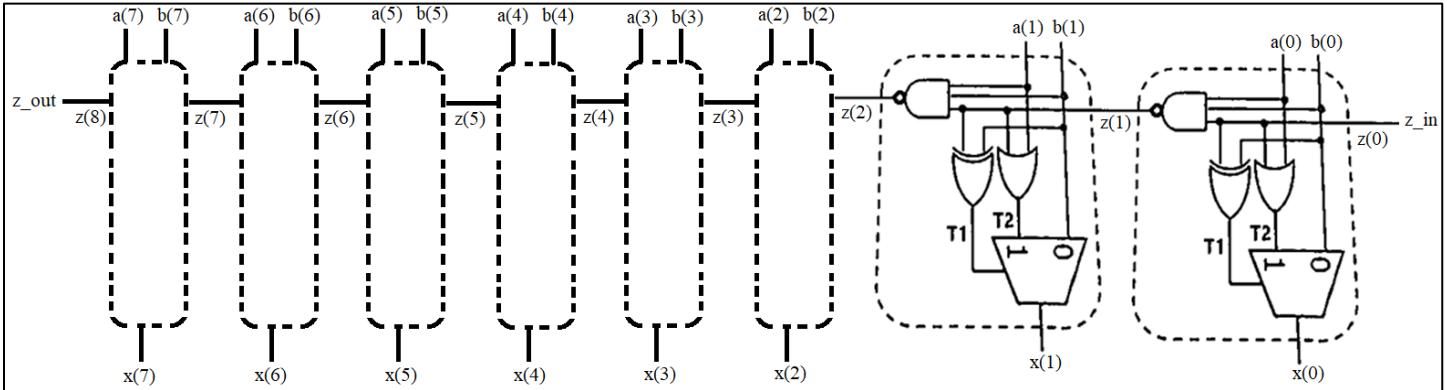
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B) Consider the below circuit named “**Circuit_logic_2**”, which consists of eight blocks of circuit named “**Circuit_logic_1**”. Write the **Structural** model VHDL code for the **Circuit_logic_2** in the below table using name association. [5pts]



Library IEEE;

use IEEE.std_logic_1164.all;

entity Circuit_logic_2 is

port (

a, b : in std_logic_vector(7 downto 0);

z_in : in std_logic;

z_out : out std_logic;

x : out std_logic_vector(7 downto 0)

);

end Circuit_logic_2;

Architecture struct_Circuit_logic_2 of Circuit_logic_2 is

component Circuit_logic_1

port (a, b, z_in : in std_logic;

z_out, x : out std_logic);

end component;

signal z : std_logic_vector(8 downto 0);

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```
begin
  z(0) <= z_in;

  GEN_LOGIC: for i in 0 to 7 generate
    LOGIC_BLOCK_i : Circuit_logic_1 port map (
      a => a(i),
      b => b(i),
      z_in => z(i),
      x => x(i),
      z_out => z(i+1)
    );
  end generate GEN_LOGIC;

  z_out <= z(8);
end struct_Circuit_logic_2;
```

تم بفضل الله،

اللهم ارحم عبدك (قاسم مفيد عمرو) واجعله من ورثة جنّة النعيم.

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